Avinash Prabhu Puppala

U.S. Green Card Holder — No Sponsorship Required

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MLE building VLMs, CNNs, and tabular models for production — from synthetic data to deployed inference at scale.

Experience

KLA | *Machine Learning & Computer Vision* — *Milpitas, CA & Austin, TX*

Senior Machine Learning Engineer — Mar. 2025 – Present

- Solved severe label scarcity (∼5 real images/class across fabs) by designing a **synthetic wafer-map generator** covering all classes — producing **100K wafer-map images** with **pixel-accurate segmentation masks and captions**.
- Trained **ConvNeXt (classification)** and **YOLO (segmentation)** models generalizing across fabs: classification **F1 +20%**, segmentation **0% → 65% mIoU**.
- Eliminated retraining on new defect classes by reframing classification as an **open-vocabulary, prompt-based task**, fine-tuning **Gemma-4 4B** via **SFT→GRPO** on synthetic wafer images: **0.80 F1** on trained classes — beating both a **31B model (0.649)** and task-specific **ConvNeXt (0.78)** — plus **0.76 F1 zero-shot** on entirely new classes.
- Ran a **vLLM serving auto-tuning study**, identifying **prefix caching (+631% throughput)** and **FP8 KV quantization (+3–19% throughput)** as highest-leverage; quantized model to **INT4 (46% size reduction, Δ−0.002 F1)**.
- Deployed with these optimizations to **214 req/s at TTFT P95 <250ms** (∼$0.008/1K requests) — **74% TTFT reduction**, **16x** over naive HuggingFace `.generate()`.

Machine Learning Engineer — Jun. 2024 – Mar. 2025

- Rebuilt legacy CPU pipeline for **E-test/WAT** data as **GPU-accelerated XGBoost**: **181x faster training**, **2.67x faster inference**, **F1 +28%**, **R^2 +32%** across four fabs (**TSMC, Micron, Samsung, Intel**). Ported to **TabICL v2**, gaining a further **∼5%** with zero fine-tuning.

Machine Learning Engineer Intern — Aug. 2023 – Dec. 2023

- Built **vision models** for low-resolution semiconductor images using self-supervised learning (**SimCLR**).
- Effectively leveraged unlabeled data, mitigating label scarcity, for a **4% AUC increase** over supervised models.
- Used **knowledge distillation** to reduce model size by **200x** while maintaining performance improvements.

Apple | *Computer Vision* — *Cupertino, CA*

Software Engineering Intern — May 2023 - Aug. 2023

- Designed a vision tool for precise **Apple HomePod LED animation detection** via video.
- **Managed project end-to-end**: conception, data pipeline, model design & training, integration into Apple's framework.
- Achieved **97% F1 score with CNN-LSTM**, significantly boosting testing efficiency and product quality across teams.

Robotics Research Center & Precog Lab, IIIT-Hyderabad | *Computer Vision, Robotics, Social Computing* — *Hyderabad, India*

Research Assistant — Sept. 2019 - Jul. 2022

- Developed a novel **CNN** to 3D reconstruct scenes from single images using **Python** & **Pytorch**. *Publication*.
- Developed a **domain-randomized** synthetic data pipeline for **Sim2Real** transfer using **C#**, **Unity** & **Blender**.
- Studied classes of users on twitter (X) using explainable ML (**SHAP values**), and models such as **XGB Classifier**, **RFC**, **LGBM** and **DNNs**. Used **SMOTE** to overcome data imbalance. *Publication 1*, *Publication 2*.
- Analysed inequities in India's COVID-19 vaccination drive using statistical tools such as **Pearson Correlation**, **Spatial Regression** and **Regression Discontinuity in Time**. *Technical Report*.

Education

Georgia Institute of Technology — Aug. 2022 - May 2024

MS in Computer Science (ML Specialization) | TA for Perception and Robotics & Deep Learning — GPA: 4.0/4.0

IIIT-Hyderabad — Jul. 2018 - Jul. 2022

B. Tech in ECE (Honors) | Dean's List for Academic & Research Excellence | TA for Computer Vision & IoT — GPA: 8.74/10

Technical Skills

Programming Languages: Python, C++, C, C#, Matlab, SQL, Bash, HTML, CSS, Javascript

Tools: Pytorch, Tensorflow, Keras, Scikit-Learn, Huggingface, Transformers, Diffusers, Numpy, Pandas, OpenCV, Unity

Coursework

ML: Statistical AI, Machine Learning, Deep Learning, Efficient Machine Learning, Natural Language Processing

Computer Vision Advanced Computer Vision, Image Processing, Mobile Robotics

Mathematics: Applied Optimization, Probability and Random Processes, Linear Algebra

Computer Science: Data Structures, Algorithms, Networks, OS, Compilers, Processor Architecture

Selected Publications

Monocular Multi-Layer Layout Estimation for Warehouse Racks	Computer Vision	ICVGIP 2021
"I'll be back": Examining Restored Accounts On Twitter	AI, Social Media	WI-IAT 2021
Efficient Rep. of Interaction Patterns with HypHC for Classification of Users on Twitter	AI, Social Media	WI-IAT 2021